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BIG DATA ANALYTICS

(LAB)

LAB FINAL REPORT

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DATE OF SUBMISSION:	13TH MAY, 2026

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Big Data Processing and Analysis of Netflix Dataset

INTRODUCTION:

In the modern digital era, organizations generate massive amounts of data that require efficient storage and processing techniques. Traditional systems are often unable to handle large-scale datasets efficiently, which has increased the importance of Big Data technologies.

This project demonstrates the implementation of Big Data analytics using the Hadoop ecosystem on a real-world Netflix dataset containing information about movies and TV shows such as title, genre, country, release year, rating, and duration. Hadoop Distributed File System (HDFS) was used for distributed data storage, Apache Spark was used for data cleaning, preprocessing, feature engineering, and advanced analytics, while Apache Hive was used to perform SQL-like queries on the processed data.

The project also includes analytical operations such as trend analysis, content segmentation, country-wise genre analysis, and director productivity analysis to generate meaningful business insights. Visualizations were created in Jupyter Notebook to better understand patterns and trends in the dataset.

The main objective of this project is to understand how Big Data tools such as Hadoop, Spark, Hive, and HBase can work together to efficiently store, process, and analyze large datasets for data-driven decision making.

Dataset Description:

The dataset used in this project is the Netflix Titles Dataset, which contains information about movies and TV shows available on the Netflix platform.

- Total Records (Rows): ~15,000+
- Total Attributes (Columns): 12 (original dataset)
- Data Type: Structured (CSV format)

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MID-TERM PART

1. HDFS SETUP

- Installed and configured Hadoop (HDFS) on Windows.
- Configured Java and Hadoop environment variables.
- Verified NameNode and DataNode services.
- Installed Apache Spark with Jupyter Notebook.
- Installed and configured Apache Hive.
- Created HDFS directories for project storage.
- Uploaded Netflix dataset to HDFS successfully.

Configuration of Hadoop:

```
C:\Users\hp>java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment Temurin-11.0.31+11 (build 11.0.31+11)
OpenJDK 64-Bit Server VM Temurin-11.0.31+11 (build 11.0.31+11, mixed mode)

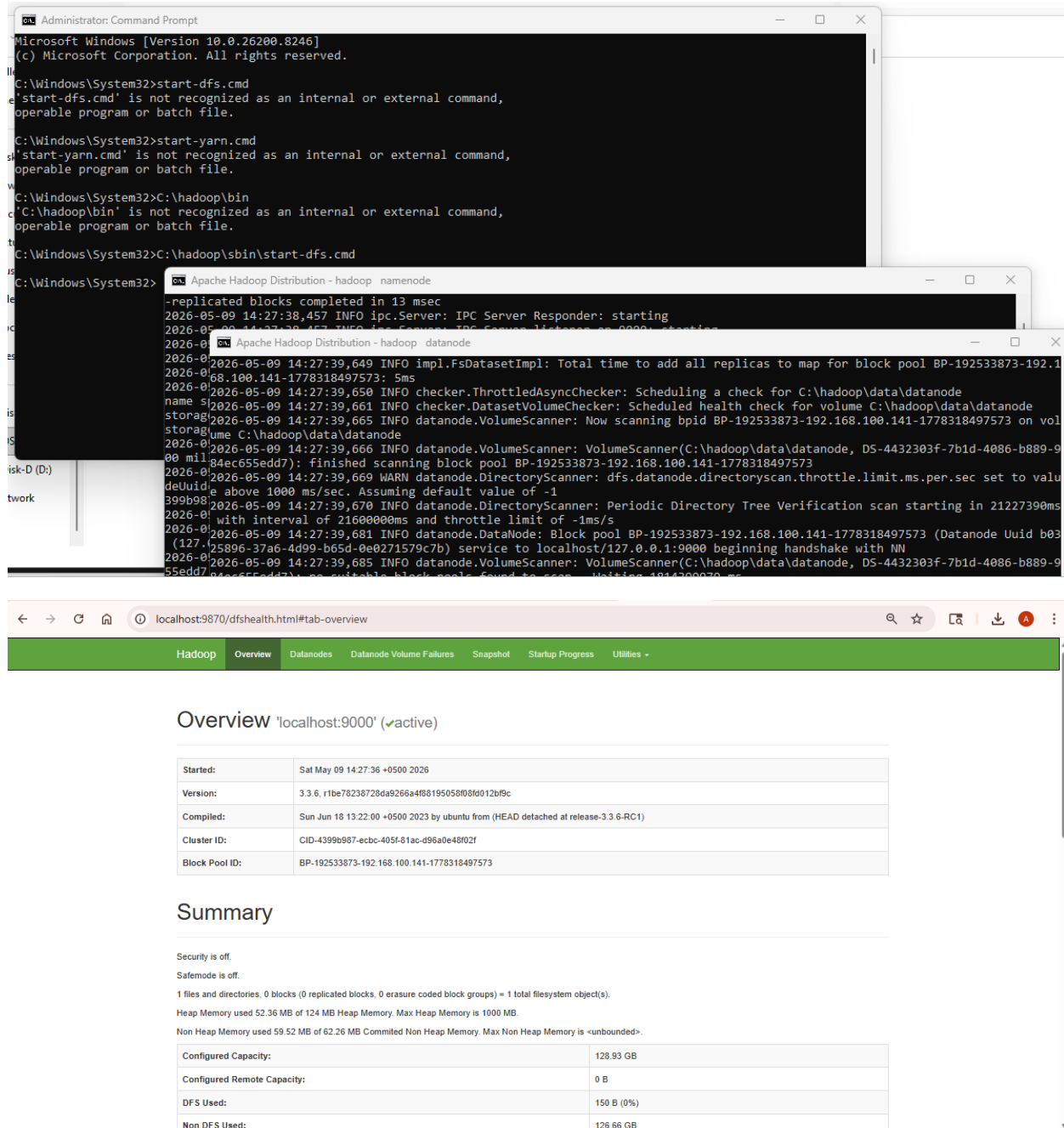
C:\Users\hp>hadoop version
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar

C:\Users\hp>
```

```
Administrator Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>hdfs namenode -format
2026-05-09 14:21:35,676 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = Com-Afshan-Lpt/192.168.100.141
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.3.6
STARTUP_MSG: classpath = C:\hadoop\etc\hadoop\hadoop\share\hadoop\common\;C:\hadoop\share\hadoop\common\lib\animal-sniffer-annotations-1.17.jar;C:\hadoop\share\hadoop\common\lib\audience-annotations-5.0.5.jar;C:\hadoop\share\hadoop\common\lib\avro-1.7.7.jar;C:\hadoop\share\hadoop\common\lib\checker-qual-2.5.2.jar;C:\hadoop\share\hadoop\common\lib\commons-beanutils-1.9.4.jar;C:\hadoop\share\hadoop\common\lib\commons-cli-1.2.jar;C:\hadoop\share\hadoop\common\lib\commons-codec-1.15.jar;C:\hadoop\share\hadoop\common\lib\commons-collections-3.2.2.jar;C:\hadoop\share\hadoop\common\lib\commons-compress-1.21.jar;C:\hadoop\share\hadoop\common\lib\commons-configuration2-2.8.0.jar;C:\hadoop\share\hadoop\common\lib\commons-daemon-1.0.13.jar;C:\hadoop\share\hadoop\common\lib\commons-io-2.8.0.jar;C:\hadoop\share\hadoop\common\lib\commons-lang3-3.12.0.jar;C:\hadoop\share\hadoop\common\lib\commons-logging-1.1.3.jar;C:\hadoop\share\hadoop\common\lib\commons-math3-3.1.1.jar;C:\hadoop\share\hadoop\common\lib\commons-net-3.9.0.jar;C:\hadoop\share\hadoop\common\lib\commons-text-1.10.0.jar;C:\hadoop\share\hadoop\common\lib\curator-client-5.2.0.jar;C:\hadoop\share\hadoop\common\lib\curator-framework-5.2.0.jar;C:\hadoop\share\hadoop\common\lib\curator-recipes-5.2.0.jar;C:\hadoop\share\hadoop\common\lib\dnsjava-2.1.7.jar;C:\hadoop\share\hadoop\common\lib\failureaccess-1.0.jar;C:\hadoop\share\hadoop\common\lib\gson-2.9.0.jar;C:\hadoop\share\hadoop\common\lib\guava-27.0-jre.jar;C:\hadoop\share\hadoop\common\lib\hadoop-annotations-3.3.6.jar;C:\hadoop\share\hadoop\common\lib\hadoop-auth-3.3.6.jar;C:\hadoop\share\hadoop\common\lib\hadoop-common\lib\hadoop-shaded-guava-1.1.1.jar;C:\hadoop\share\hadoop\common\lib\hadoop-shaded-protobuf-3_7-1.1.1.jar;C:\hadoop\share\hadoop\common\lib\httpclient-4.5.13.jar;C:\hadoop\share\hadoop\common\lib\httpcore-4.4.13.jar;C:\hadoop\share\hadoop\common\lib\j2objc-annotations-1.1.jar;C:\hadoop\share\hadoop\common\lib\jackson-annotations-2.12.7.jar;C:\hadoop\share\hadoop\common\lib\jackson-core-2.12.7.jar;C:\hadoop\share\hadoop\common\lib\jackson-core-asl-1.9.13.jar;C:\hadoop\share\hadoop\common\lib\jackson-databind-2.12.7.1.jar;C:\hadoop\share\hadoop\common\lib\jackson-mapper-asl-1.9.13.jar;C:\hadoop\share\hadoop\common\lib\jakarta.activation-api-1.2.1.jar;C:\hadoop\share\hadoop\common\lib\javax.servlet-api-3.1.0.jar;C:\hadoop\share\hadoop\common\lib\jaxb-api-2.2.11.jar;C:\hadoop\share\hadoop\common\lib\jaxb-impl-2.2.3-1.jar;C:\hadoop\share\hadoop\common\lib\jcip-annotations-1.0-1.jar;C:\hadoop\share\hadoop\common\lib\jersey-core-1.19.4.jar;C:\hadoop\share\hadoop\common\lib\jersey-jackson-1.19.4.jar;C:\hadoop\share\hadoop\common\lib\jersey-server-1.19.4.jar;C:\hadoop\share\hadoop\common\lib\jetty-http-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-io-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-security-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-server-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-servlet-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-util-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jetty-webapp-9.4.51.v20230217.jar;C:\hadoop\share\hadoop\common\lib\jsr305-3.0.2.jar;C:\hadoop\share\hadoop\common\lib\jsr311-api-1.1.1.jar;C:\hadoop\share\hadoop\common\lib\jul-to-slf4j-1.7.36.jar;C:\hadoop\share\hadoop\common\lib\kerb-admin-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-client-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-common-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-identity-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-logging-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-simplekdc-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerb-util-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerby-asn1-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerby-config-1.0.1.jar;C:\hadoop\share\hadoop\common\lib\kerby-pkix-1.0.1.jar;C:\hadoop\share\hadoop\
*****/
```

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The image shows a Windows command prompt and the Hadoop web interface. The command prompt shows the execution of `start-dfs.cmd`, `start-yarn.cmd`, and `start-dfs.cmd` in the `C:\hadoop\bin` directory. The Hadoop web interface shows the 'Overview' tab for 'localhost:9000' (active). The dashboard includes a table with Hadoop configuration details and a summary section.

Overview 'localhost:9000' (✓active)

Started:	Sat May 09 14:27:36 +0500 2026
Version:	3.3.6, r1be78238728da9266a4f881950580d012b9c
Compiled:	Sun Jun 18 13:22:00 +0500 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID:	CID-4399b987-ecbc-405f-81ac-d96a0e4802f
Block Pool ID:	BP-192533873-192.168.100.141-1778318497573

Summary

Security is off.
Safemode is off.
1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).
Heap Memory used 52.36 MB of 124 MB Heap Memory. Max Heap Memory is 1000 MB.
Non Heap Memory used 59.52 MB of 62.26 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	128.93 GB
Configured Remote Capacity:	0 B
DFS Used:	150 B (0%)
Non DFS Used:	126.66 GB

[illegible]

The screenshot shows the Hadoop cluster management interface. On the left is a navigation menu with links: About, Nodes, Node Labels, Applications, NEW, NEW SAVING, SUBMITTED, ACCEPTED, RUNNING, FINISHED, FAILED, KILLED, Scheduler, and Tools. The main content area is titled 'All Applications' and displays 'Cluster Metrics' and 'Cluster Nodes Metrics'. The 'Cluster Metrics' table shows 0 Apps Submitted, 0 Apps Pending, 0 Apps Running, 0 Apps Completed, 0 Containers Running, and Used Resources of <memory:0 B, vCores:0>. The 'Cluster Nodes Metrics' table shows 0 Active Nodes, 0 Decommissioning Nodes, 0 Decommissioned Nodes, and 1 Lost Nodes. Below this is the 'Scheduler Metrics' section, which includes a table for Capacity Scheduler showing Minimum Allocation of <memory:1024, vCores:1> and Maximum Allocation of <memory:8192, vCores:4>. At the bottom, there is a table for 'Show 20 entries' with columns: ID, User, Name, Application Type, Application Tags, Queue, Application Priority, StartTime, LaunchTime, FinishTime, State, FinalStatus, Running Containers, Allocated CPU vCores, and Allocated Memory MB. A message at the bottom states 'Showing 0 to 0 of 0 entries'.

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used Resources	Total Resources
0	0	0	0	0	<memory:0 B, vCores:0>	<memory:0 B, vCores:0>

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes
0	0	0	1

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU vCores	Allocated Memory MB
No data available in table														

Showing 0 to 0 of 0 entries

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>hdfs dfs -mkdir /project

C:\Users\hp>hdfs dfs -ls /
Found 1 items
drwxr-xr-x  - hp supergroup          0 2026-05-09 14:35 /project
```

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Creating Spark Session:

```
jupyter BIG DATA LAB FINAL Last Checkpoint: 1 hour ago
File Edit View Run Kernel Settings Help
+ ✂ 📄 📄 ▶ ■ 🔁 ⏩ Code ▾

[ ]: #SP23-BDS-039 NAYAB KHALID #SP23-BDS-040 NOOR UL HUDA BATOOL

[ ]: #BIG DATA LAB PROJECT

[ ]: #MIDTERM PART

[2]: from pyspark.sql import SparkSession

    spark = SparkSession.builder \
        .appName("NetflixBigDataProject") \
        .master("local[*]") \
        .config("spark.sql.warehouse.dir", "hdfs://localhost:9000/user/hive/warehouse") \
        .enableHiveSupport() \
        .getOrCreate()

[3]: spark.sql("SHOW DATABASES").show()

+-----+
|namespace|
+-----+
| default|
+-----+
```

Uploading Netflix dataset:

```
jupyter BIG DATA LAB FINAL Last Checkpoint: 1 hour ago Trusted
File Edit View Run Kernel Settings Help
+ ✂ 📄 📄 ▶ ■ 🔁 ⏩ Code ▾ JupyterLab Python 3 (ipykernel)

[4]: df = spark.read.csv(
    "hdfs://localhost:9000/netflix_project/netflix_titles.csv",
    header=True,
    inferSchema=True
)

[5]: df.show(5)

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|show_id|type|title|director|cast|country|date_added|release_year|rating|duration|list
ed_in|description|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|s1|Movie|Dick Johnson Is Dead|Kirsten Johnson|NULL|United States|September 25, 2021|2020|PG-13|90 min|Document
aries|As her father nea...
|s2|TV Show|Blood & Water|NULL|Ama Qamata, Khosi...|South Africa|September 24, 2021|2021|TV-MA|2 Seasons|International T
V ...|After crossing pa...
|s3|TV Show|Ganglands|Julien Leclercq|Sami Bouajila, Tr...|NULL|September 24, 2021|2021|TV-MA|1 Season|Crime TV Shows,
I...|To protect his fa...
|s4|TV Show|Jailbirds New Orl...|NULL|NULL|NULL|September 24, 2021|2021|TV-MA|1 Season|Docuseries, Rea
li...|Feuds, flirtation...
|s5|TV Show|Kota Factory|NULL|Mayur More, Jiten...|India|September 24, 2021|2021|TV-MA|2 Seasons|International T
V ...|In a city of coac...
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
```

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```
[6]: df.printSchema()
```

```
root
|-- show_id: string (nullable = true)
|-- type: string (nullable = true)
|-- title: string (nullable = true)
|-- director: string (nullable = true)
|-- cast: string (nullable = true)
|-- country: string (nullable = true)
|-- date_added: string (nullable = true)
|-- release_year: string (nullable = true)
|-- rating: string (nullable = true)
|-- duration: string (nullable = true)
|-- listed_in: string (nullable = true)
|-- description: string (nullable = true)
```

Command Prompt Windows PowerShell Windows PowerShell Windows PowerShell

Windows PowerShell
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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\hp> hdfs dfs -mkdir /netflix_project
PS C:\Users\hp> hdfs dfs -put C:\BigDataProject\netflix_titles.csv /netflix_project
PS C:\Users\hp> hdfs dfs -ls /netflix_project
Found 1 items
-rw-r--r--  1 hp supergroup    3399671 2026-05-09 15:27 /netflix_project/netflix_titles.csv
PS C:\Users\hp>
```

BigDataProject

< > ↑ ↺ > This PC > OS (C:) > BigDataProject

New ✂ 📄 📁 📧 📧 🗑 Sort View ...

	Name	Date modified	Type	Size
	netflix_titles	5/9/2026 12:54 PM	Microsoft Excel C...	3,320 KB

Home Gallery OneDrive - Persi

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2. DATA CLEANING USING SPARK

➤ Cleaning the dataset:

Data Preprocessing (Using Spark):

The dataset was cleaned and transformed using PySpark:

- Loaded dataset from HDFS
- Removed duplicate records
- Handled missing values (filled with “Unknown”)
- Converted and cleaned date fields
- Extracted new features:
- Year added
- Month added
- Duration numeric values
- Standardized text fields (lowercase conversion)
- Created additional columns such as content type indicators
- Saved cleaned dataset back to HDFS in Parquet format

➤ Handling Missing & Duplicate Values:

```
[9]: #CLEAN NULL VALUES

df = df.fillna({
    "director": "Unknown",
    "cast": "Unknown",
    "country": "Unknown",
    "rating": "Not Rated"
})
```

```
[10]: #REMOVE DUPLICATES
df = df.dropDuplicates()
```

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➤ Adding derived Fields:

```
•[9]: #ADD DERIVED COLUMNS
      from pyspark.sql.functions import *

•[10]: #Convert date:
      df = df.withColumn(
          "date_added",
          to_date(col("date_added"), "MMMM d, yyyy")
      )

•[11]: #Create year/month:
      df = df.withColumn(
          "added_year",
          year("date_added")
      )

      df = df.withColumn(
          "added_month",
          month("date_added")
      )

•[12]: #Create content age:
      df = df.withColumn(
          "content_age",
          year(current_date()) - col("release_year")
      )
```

➤ Cleaned data:

```
[13]: df.show(5)
```

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	
isted_in		description	added_year	added_month	content_age					1
s122	TV Show	Hotel Del Luna	Unknown	Lee Ji-eun, Yeo J...	Unknown	2021-09-02	2019	TV-14	1 Season	Internationa
l TV ...	When he's invited...	2021	9	7.0						
s461	Movie	Surf's Up	Ash Brannon, Chri...	Shia LaBeouf, Jef...	United States, Ca...	2021-07-15	2007	PG	86 min	Children & F
amily...	This Oscar-nomina...	2021	7	19.0						
s660	TV Show	Bangkok Love Stor...	Unknown	Nida Patcharaveer...	Thailand	2021-06-19	2018	TV-14	1 Season	Internationa
l TV ...	From a teenage pa...	2021	6	8.0						
s695	Movie	Aziza	Soudade Kaadan	Careess Bashar, Ab...	Lebanon, Syria	2021-06-17	2019	TV-PG	13 min	Comedies, Dr
amas,...	This short film f...	2021	6	7.0						
s883	Movie	Jungle Beat: The ...	Brent Dawes	Ed Kear, David Me...	Mauritius	2021-05-14	2021	TV-Y7	90 min	Children & F
amily...	When a lost and l...	2021	5	5.0						

only showing top 5 rows

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3. STORE CLEAN DATA

HDFS Path:

➤ Save Cleaned Data To HDFS:

```
•[17]: #SAVE CLEAN DATA TO HDFS

df.write.mode("overwrite").parquet(
    "hdfs://localhost:9000/netflix_project/cleaned_data"
)
```

Verifying Cleaned Data in HDFS:

```
PS C:\Users\hp> hdfs dfs -ls /netflix_project/cleaned_data
Found 4 items
-rw-r--r--  3 hp supergroup          0 2026-05-09 15:50 /netflix_project/cleaned_data/_SUCCESS
-rw-r--r--  3 hp supergroup    684314 2026-05-09 15:50 /netflix_project/cleaned_data/part-00000-d48f5b62-53be-4e21-ab1
8-d48e441b1ed8-c000.snappy.parquet
-rw-r--r--  3 hp supergroup    681568 2026-05-09 15:50 /netflix_project/cleaned_data/part-00001-d48f5b62-53be-4e21-ab1
8-d48e441b1ed8-c000.snappy.parquet
-rw-r--r--  3 hp supergroup    702416 2026-05-09 15:50 /netflix_project/cleaned_data/part-00002-d48f5b62-53be-4e21-ab1
8-d48e441b1ed8-c000.snappy.parquet
PS C:\Users\hp>
```

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4. HIVE QUERIES

Configuration of Hive:

```
Command Prompt
C:\Users\hp>hdfs dfs -mkdir /user
C:\Users\hp>hdfs dfs -mkdir /user/hive
C:\Users\hp>hdfs dfs -mkdir /user/hive/warehouse
C:\Users\hp>hdfs dfs -mkdir /user
mkdir: `/user': File exists

C:\hive\bin>dir
Volume in drive C is OS
Volume Serial Number is 6ACC-B930

Directory of C:\hive\bin

05/09/2026 03:14 PM <DIR>      .
05/09/2026 03:14 PM <DIR>      ..
10/24/2019 08:17 AM             881 beeline
05/09/2026 03:14 PM <DIR>      ext
03/29/2022 12:39 AM          10,158 hive
02/28/2022 03:31 AM          2,085 hive-config.sh
10/24/2019 08:17 AM             885 hiveserver2
10/24/2019 08:17 AM             880 hplsql
10/24/2019 08:17 AM          3,064 init-hive-dfs.sh
10/24/2019 08:17 AM             832 metatool
10/24/2019 08:17 AM             884 schematool
                        8 File(s)          19,669 bytes
                        3 Dir(s)          441,950,208 bytes free

C:\hive\bin>
```

➤ Creating external Hive tables mapped to this data:

```
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[34]: spark.sql("DROP TABLE IF EXISTS netflix_data")

[34]: DataFrame[]

*[35]: #CREATE HIVE TABLE
spark.sql("""
CREATE EXTERNAL TABLE netflix_data (
  show_id STRING,
  type STRING,
  title STRING,
  director STRING,
  cast STRING,
  country STRING,
  date_added DATE,
  release_year STRING,
  rating STRING,
  duration STRING,
  listed_in STRING,
  description STRING,
  added_year INT,
  added_month INT,
  content_age INT
)
STORED AS PARQUET
LOCATION 'hdfs://localhost:9000/netflix_project/cleaned_data'
""")

[35]: DataFrame[]
```

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- Verifying Data

```
•[36]: #VERIFY TABLE
spark.sql("SHOW TABLES").show()

+-----+-----+-----+
|namespace|tableName|isTemporary|
+-----+-----+-----+
| default|netflix_data| false|
+-----+-----+-----+
```

- Count Movies vs TV Shows

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 Code ▾

[1]: #HIVE QUERIES

[37]: #QUERY 1 – Count Movies vs TV Shows

```
spark.sql("""
SELECT type, COUNT(*) as total_content
FROM netflix_data
GROUP BY type
""").show()
```

```
+-----+-----+
|          type|total_content|
+-----+-----+
|          NULL|            1|
|        TV Show|          2676|
|         Movie|          6131|
|William Wyler|            1|
+-----+-----+
```

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- Top Categories

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```
[38]: #QUERY 2 - Top Categories
      spark.sql("""
      SELECT listed_in, COUNT(*) as total
      FROM netflix_data
      GROUP BY listed_in
      ORDER BY total DESC
      LIMIT 10
      """).show()
```

```
+-----+-----+
|          listed_in|total|
+-----+-----+
|Dramas, Internati...| 361|
|      Documentaries| 358|
|      Stand-Up Comedy| 334|
|Comedies, Dramas,...| 273|
|Dramas, Independe...| 252|
|          Kids' TV| 220|
|Children & Family...| 215|
|Children & Family...| 201|
|Documentaries, In...| 186|
|Dramas, Internati...| 180|
+-----+-----+
```

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- Filter Recent Content

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File Edit View Run Kernel Settings Help

Code

```
[39]: #QUERY 3 - Filter Recent Content
Recent Contentspark.sql("""
spark.sql("""
SELECT title, release_year
FROM netflix_data
WHERE release_year >= 2020
""").show()
```

title	release_year
The Paper Tigers	2021
Car Masters: Rust...	2021
My Unorthodox Life	2021
Mama Drama	2020
Super Monsters: O...	2021
Tuesdays & Fridays	2021
Age of Samurai: B...	2021
Pelé	2021
Pretend It's a City	2021
Dream Home Makeover	2021
Monarca	2021
Natalie Palamides...	2020
The Trials of Gab...	2020
Ghost Stories	2020
Carnaval	2021
I Am All Girls	2021
Reframe THEATER E...	2020
Mr. Iglesias	2020
Hoops	2020
All the Freckles ...	2020

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- Top 10 Countries with Most Netflix Content

```
[41]: #QUERY 4 – Top 10 Countries with Most Netflix Content
spark.sql("""
SELECT country, COUNT(*) AS total_content
FROM netflix_data
WHERE country IS NOT NULL
GROUP BY country
ORDER BY total_content DESC
LIMIT 10
""").show()
```

country	total_content
United States	2805
India	972
Unknown	832
United Kingdom	419
Japan	245
South Korea	199
Canada	181
Spain	145
France	123
Mexico	110

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- Number of Movies and TV Shows Released Each Year

```
[42]: #QUERY 5 – Number of Movies and TV Shows Released Each Year
spark.sql("""
SELECT release_year, type, COUNT(*) AS total_titles
FROM netflix_data
GROUP BY release_year, type
ORDER BY release_year DESC
""").show()
```

release_year	type	total_titles
United States	Movie	1
June 12, 2021	Movie	1
January 15, 2021	Movie	1
January 13, 2021	Movie	1
December 15, 2020	TV Show	1
August 13, 2020	Movie	1
40 min William Wyler		1
2021	Movie	275
2021	TV Show	314
2020	TV Show	436
2020	Movie	516
2019	Movie	630
2019	TV Show	396
2018	Movie	765
2018	TV Show	380
2017	TV Show	265
2017	Movie	765
2016	TV Show	244
2016	Movie	657
2015	Movie	397

FINDINGS:

Using Hive queries, several analytical insights were obtained from the Netflix dataset:

- Analysis of Movies and TV Shows distribution
- Identification of top countries contributing Netflix content
- Filtering content based on release year and time trends
- Analysis of genres and ratings distribution
- Retrieval of recent and high-volume content information

These findings demonstrate how Hive can be used to perform SQL-like analytical operations on large-scale datasets to generate meaningful business insights.

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FINAL-TERM PART

Importing Libraries & Functions:

```
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[43]: #FINAL PART

•[44]: #VERIFYING IF TABLE EXISTS
spark.sql("SHOW TABLES").show()

+-----+-----+
|namespace|  tableName|isTemporary|
+-----+-----+
| default|netflix_data|      false|
+-----+-----+

•[46]: #IMPORTING REQUIRED LIBRARIES
from pyspark.sql.functions import *
from pyspark.sql.window import Window

import matplotlib.pyplot as plt
import pandas as pd
```

1. ADVANCED ANALYTICS IN SPARK

- Yearly Content Growth Analysis:

```
[ ]: # Advanced Analytics in Spark

•[47]: # 1. YEARLY CONTENT GROWTH ANALYSIS

[48]: growth_df = spark.sql("""
SELECT
  CAST(release_year AS INT) AS year,
  COUNT(*) AS total_titles
FROM netflix_data
GROUP BY release_year
ORDER BY year
""")

[49]: windowSpec = Window.orderBy("year")

[50]: growth_df = growth_df.withColumn(
  "previous_year",
  lag("total_titles").over(windowSpec)
)

[51]: growth_df = growth_df.withColumn(
  "growth_rate",
  round(
    (
      (col("total_titles") - col("previous_year"))
      / col("previous_year")
    ) * 100,
    2
  )
)
```

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```
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+ + + + + Code
[52]: growth_df.show()

+-----+
|year|total_titles|previous_year|growth_rate|
+-----+
|NULL|1|NULL|NULL|
|NULL|1|1|0.0|
|NULL|2|1|100.0|
|NULL|1|2|-50.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
|NULL|1|1|0.0|
+-----+
only showing top 20 rows
```

- **Content Segmentation** (behavior grouping)

```
[53]: # 2. CONTENT SEGMENTATION
      # (behavior grouping)

[54]: content_segments = df.withColumn(
      "content_segment",
      when(col("content_age") <= 5, "Recent")
      .when(col("content_age") <= 15, "Modern")
      .otherwise("Classic")
    )

[55]: segment_counts = content_segments.groupBy(
      "content_segment"
    ).count()

[56]: segment_counts.show()

+-----+
|content_segment|count|
+-----+
|Modern|6675|
|Recent|589|
|Classic|1545|
+-----+
```

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- **Monthly Content Addition Trend** (time-based aggregation)

```
•[57]: # 3. MONTHLY CONTENT ADDITION TREND
      # (time-based aggregation)
```

```
[58]: monthly_trend = spark.sql("""
      SELECT
      added_month,
      COUNT(*) AS total_added
      FROM netflix_data
      GROUP BY added_month
      ORDER BY added_month
      """)
```

```
[59]: monthly_trend.show()
```

```
+-----+-----+
|added_month|total_added|
+-----+-----+
|      NULL|         120|
|         1|         722|
|         2|         557|
|         3|         732|
|         4|         757|
|         5|         625|
|         6|         723|
|         7|         818|
|         8|         746|
|         9|         763|
|        10|         754|
|        11|         696|
|        12|         796|
+-----+-----+
```

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- Country + Genre Analysis (Category Level)

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```
*[60]: # 4. COUNTRY + GENRE ANALYSIS
      # (Category Level)
```

```
[61]: country_genre = spark.sql("""
      SELECT
      country,
      listed_in,
      COUNT(*) AS total
      FROM netflix_data
      WHERE country IS NOT NULL
      GROUP BY country, listed_in
      ORDER BY total DESC
      """)
```

```
[62]: country_genre.show(20, truncate=False)
```

country	listed_in	total
United States	Documentaries	248
United States	Stand-Up Comedy	209
India	Comedies, Dramas, International Movies	120
India	Dramas, International Movies	118
India	Dramas, Independent Movies, International Movies	108
United States	Children & Family Movies, Comedies	90
United States	Dramas	88
United States	Comedies	84
United States	Children & Family Movies	80
United States	Kids' TV	77
Japan	Anime Series, International TV Shows	75
United States	Dramas, Independent Movies	72
Unknown	Children & Family Movies	70
United States	Reality TV	70
United States	Action & Adventure	64
United States	Comedies, Dramas, Independent Movies	63
South Korea	International TV Shows, Korean TV Shows, Romantic TV Shows	63
India	Dramas, International Movies, Romantic Movies	62
United States	Docuseries	61

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- **Director Productivity Analysis**

```
[63]: #DIRECTOR PRODUCTIVITY ANALYSIS
```

```
[64]: director_analysis = spark.sql("""
      SELECT
      director,
      COUNT(*) AS total_titles
      FROM netflix_data
      WHERE director != 'Unknown'
      GROUP BY director
      ORDER BY total_titles DESC
      LIMIT 15
      """)
```

```
[65]: director_analysis.show(truncate=False)
```

director	total_titles
Rajiv Chilaka	19
Raúl Campos, Jan Suter	18
Marcus Raboy	16
Suhas Kadav	16
Jay Karas	14
Cathy Garcia-Molina	13
Youssef Chahine	12
Martin Scorsese	12
Jay Chapman	12
Steven Spielberg	11
Don Michael Paul	10
David Dhawan	9
Robert Rodriguez	8
Troy Miller	8
Kunle Afolayan	8

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2. STORE KEY RESULTS IN HBASE

- Starting Hbase:

```
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•[66]: # HBASE SECTION

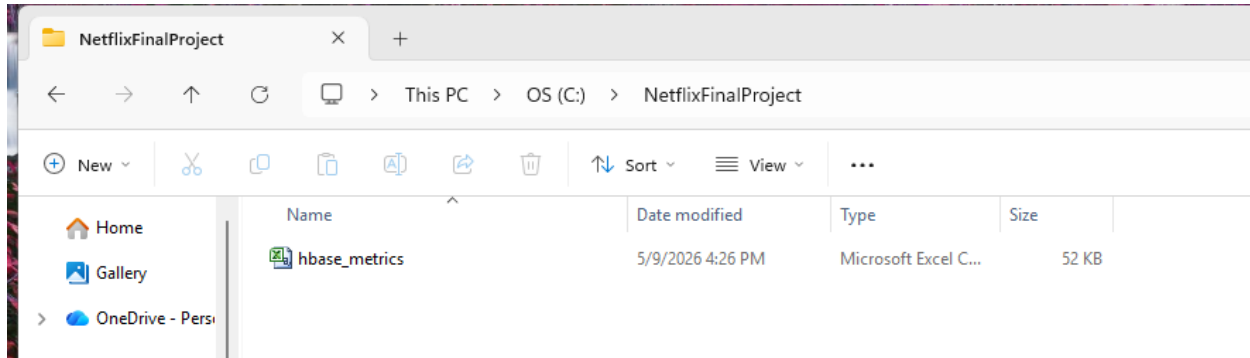
•[67]: #CREATING HBASE METRICS DATAFRAME
hbase_metrics = spark.sql("""
SELECT
country,
COUNT(*) AS total_titles,
MAX(listed_in) AS dominant_genre
FROM netflix_data
GROUP BY country
ORDER BY total_titles DESC
""")

[68]: hbase_metrics.show()

+-----+-----+-----+
|          country|total_titles|    dominant_genre|
+-----+-----+-----+
|      United States|        2805|          Thrillers|
|             India|         972|          TV Shows|
|          Unknown|         832|          Thrillers|
|    United Kingdom|         419|          Thrillers|
|             Japan|         245|          TV Shows|
|    South Korea|         199|    Stand-Up Comedy|
|          Canada|         181|          Thrillers|
|            Spain|         145|          Thrillers|
|          France|         123|    Stand-Up Comedy|
|          Mexico|         110|    Stand-Up Comedy|
|          Egypt|         106|International TV ...|
|          Turkey|         105|International TV ...|
|          Nigeria|          93|International TV ...|
|        Australia|          87|          Thrillers|
|          Taiwan|          81|International TV ...|
|        Indonesia|          79|International TV ...|
|          Brazil|          77|    Stand-Up Comedy|
|    Philippines|          75|International TV ...|
|United Kingdom, U...|          75|          Thrillers|
|United States, Ca...|          73|TV Horror, TV Mys...|
+-----+-----+-----+

•[70]: #EXPORT HBASE METRICS
hbase_metrics.toPandas().to_csv(
    "C:/NetflixFinalProject/hbase_metrics.csv",
    index=False
)
```

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3. JOIN HIVE + HBASE

Spark was used to join Hive data with HBase-style summarized metrics to perform enriched analytics and integrated data analysis.

```
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•[71]: # Join Hive + HBase

[72]: hbase_df = spark.read.csv(
      "C:/NetflixFinalProject/hbase_metrics.csv",
      header=True,
      inferSchema=True
    )

[73]: hbase_df.show()

+-----+-----+-----+
|country|total_titles|dominant_genre|
+-----+-----+-----+
|United States|2805|Thrillers|
|India|972|TV Shows|
|Unknown|832|Thrillers|
|United Kingdom|419|Thrillers|
|Japan|245|TV Shows|
|South Korea|199|Stand-Up Comedy|
|Canada|181|Thrillers|
|Spain|145|Thrillers|
|France|123|Stand-Up Comedy|
|Mexico|110|Stand-Up Comedy|
|Egypt|106|International TV ...|
|Turkey|105|International TV ...|
|Nigeria|93|International TV ...|
|Australia|87|Thrillers|
|Taiwan|81|International TV ...|
|Indonesia|79|International TV ...|
|Brazil|77|Stand-Up Comedy|
|Philippines|75|International TV ...|
|United Kingdom, U...|75|Thrillers|
|United States, Ca...|73|TV Horror, TV Mys...|
+-----+-----+-----+
only showing top 20 rows
```

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```
[74]: hbase_df.createOrReplaceTempView("hbase_metrics")
```

```
[75]: joined_df = spark.sql("""
SELECT
n.title,
n.country,
n.rating,
h.total_titles,
h.dominant_genre
FROM netflix_data n
JOIN hbase_metrics h
ON n.country = h.country
""")
```

```
•[76]: #VIEW JOINED RESULT
joined_df.show()
```

title	country	rating	total_titles	dominant_genre
The Paper Tigers	United States	PG-13	2805	Thrillers
Car Masters: Rust...	Unknown	TV-PG	832	Thrillers
Chhota Bheem & Kr...	Unknown	TV-Y7	832	Thrillers
My Unorthodox Life	Unknown	TV-MA	832	Thrillers
Mama Drama	Unknown	TV-MA	832	Thrillers
A Haunted House 2	United States	R	2805	Thrillers
Super Monsters: O...	Unknown	TV-Y	832	Thrillers
Legal Hash	Unknown	TV-MA	832	Thrillers
Tuesdays & Fridays	India	TV-14	972	TV Shows
Age of Samurai: B...	United States	TV-MA	2805	Thrillers
Pelé	United Kingdom	TV-14	419	Thrillers
Pretend It's a City	United States	TV-14	2805	Thrillers
Dream Home Makeover	United States	TV-G	2805	Thrillers
Monarca	Mexico	TV-MA	110	Stand-Up Comedy
Unknown	United Kingdom, G...	PG-13	1	Action & Adventur...
Natalie Palamides...	United States	TV-MA	2805	Thrillers
The Trials of Gab...	United States	TV-MA	2805	Thrillers
Ghost Stories	India	TV-MA	972	TV Shows

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4. VISUALIZATIONS

- Growth Rate Through Line Chart:

```
•[77]: # VISUALIZATIONS

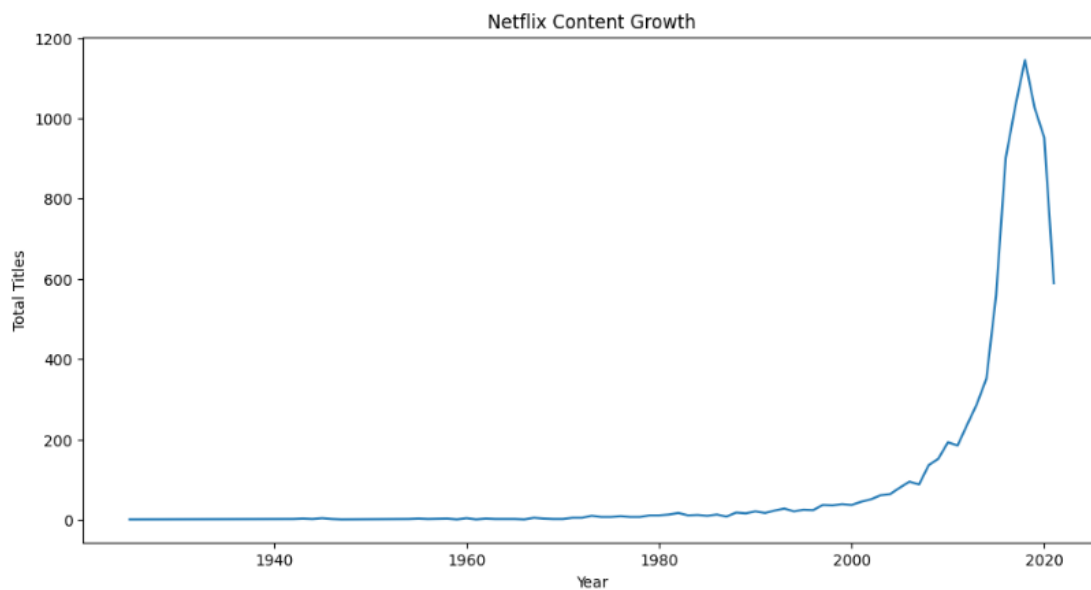
•[78]: # 1. GROWTH RATE THROUGH LINE CHART
growth_pd = growth_df.toPandas()

plt.figure(figsize=(12,6))

plt.plot(
    growth_pd["year"],
    growth_pd["total_titles"]
)

plt.title("Netflix Content Growth")
plt.xlabel("Year")
plt.ylabel("Total Titles")

plt.show()
```



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- Content Segments Through Pie Chart:

```
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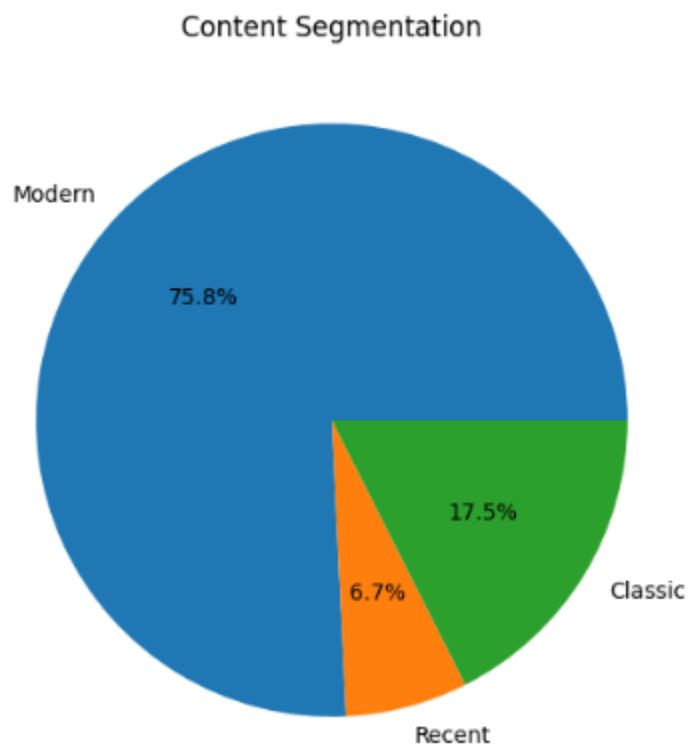
•[79]: # 2. CONTENT SEGMENTS THROUGH PIE CHART
segment_pd = segment_counts.toPandas()

plt.figure(figsize=(6,6))

plt.pie(
    segment_pd["count"],
    labels=segment_pd["content_segment"],
    autopct='%1.1f%%'
)

plt.title("Content Segmentation")

plt.show()
```



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- Monthly Trend Through Line Chart:

```
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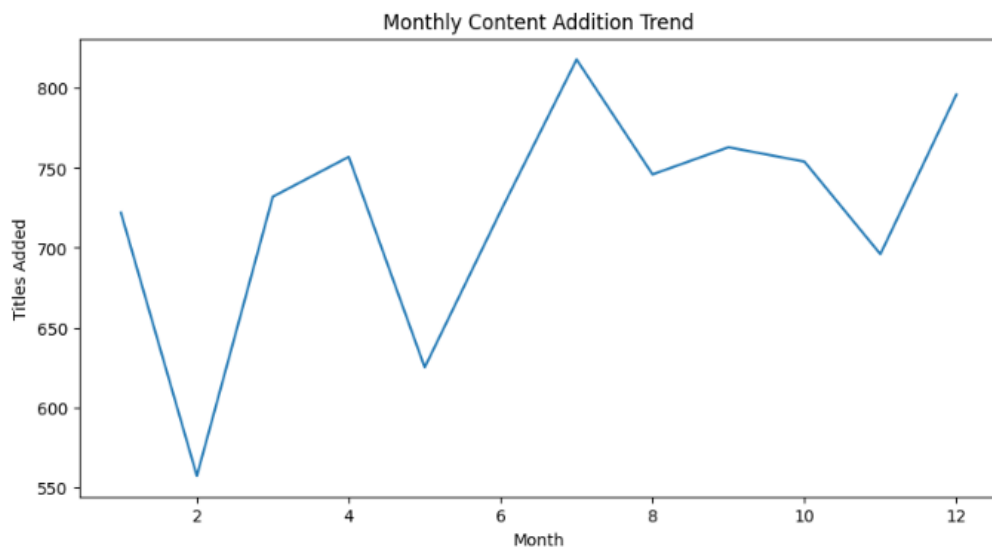
•[80]: # 3. MONTHLY TREND THROUGH LINE CHART
monthly_pd = monthly_trend.toPandas()

plt.figure(figsize=(10,5))

plt.plot(
    monthly_pd["added_month"],
    monthly_pd["total_added"]
)

plt.title("Monthly Content Addition Trend")
plt.xlabel("Month")
plt.ylabel("Titles Added")

plt.show()
```



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- Director Analysis Though Horizontal Bar Chart:

```
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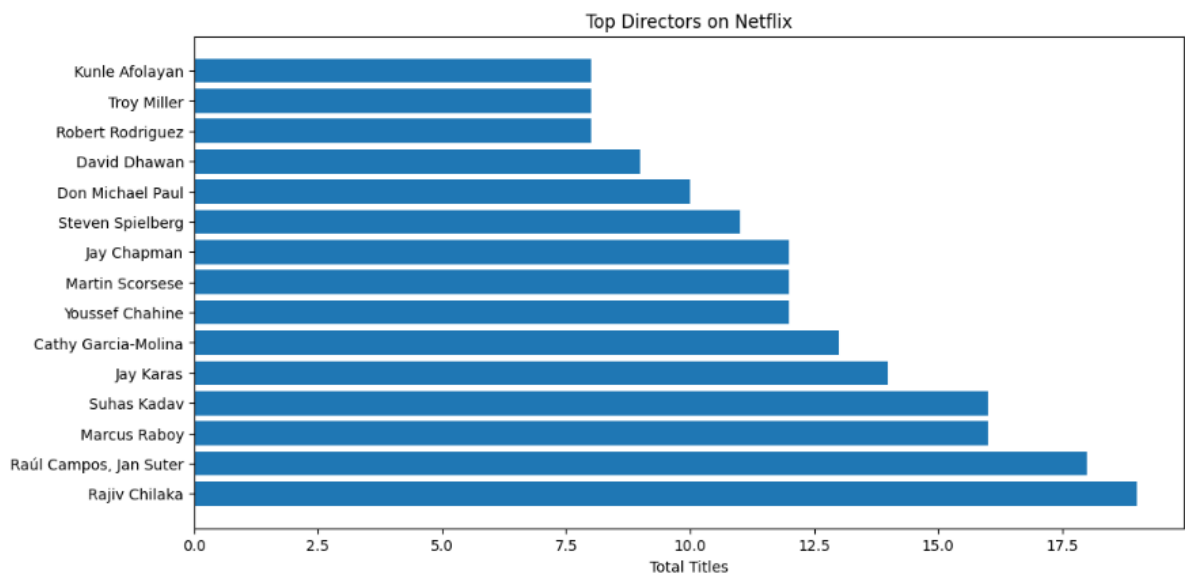
•[88]: # 4. DIRECTOR ANALYSIS THOUGH HORIZONTAL BAR CHART
director_pd = director_analysis.toPandas()

plt.figure(figsize=(12,6))

plt.barh(
    director_pd["director"],
    director_pd["total_titles"]
)

plt.title("Top Directors on Netflix")
plt.xlabel("Total Titles")

plt.show()
```



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- Ratings Distribution Chart Through Bar Chart:**

```
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•[89]: # 5. RATINGS DISTRIBUTION CHART THROUGH BAR CHART

ratings = spark.sql("""
SELECT
rating,
COUNT(*) AS total
FROM netflix_data
GROUP BY rating
ORDER BY total DESC
""")
ratings_pd = ratings.toPandas()

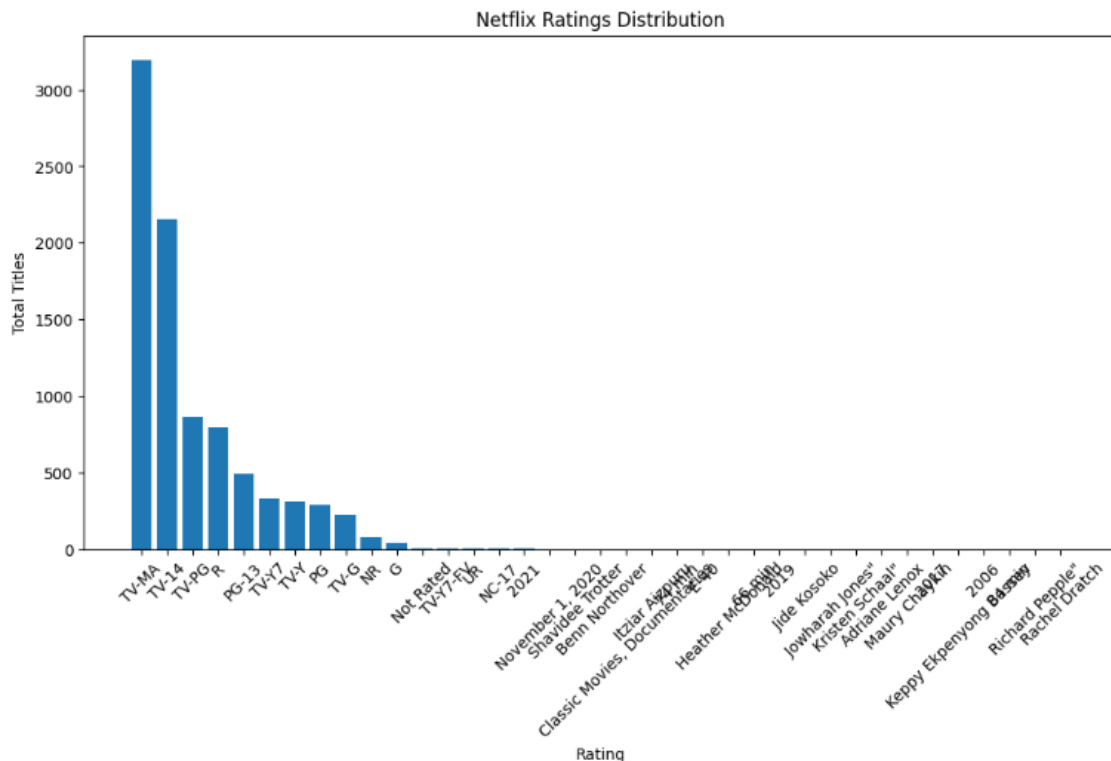
plt.figure(figsize=(12,6))

plt.bar(
    ratings_pd["rating"],
    ratings_pd["total"]
)

plt.title("Netflix Ratings Distribution")
plt.xlabel("Rating")
plt.ylabel("Total Titles")

plt.xticks(rotation=45)

plt.show()
```



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5. ANALYSIS REPORT

Key Insights:

- Netflix content has grown significantly over the years, especially after 2015, indicating rapid platform expansion.
- Movies constitute a larger portion of the platform compared to TV Shows.
- Certain countries such as the United States and India contribute the highest amount of content.
- Drama, International Movies, and Comedy-related genres are among the most dominant categories on the platform.
- Recent and modern content dominate the dataset, showing Netflix's focus on continuously updating its content library.
- Some directors contribute a significantly higher number of titles, indicating strong creative partnerships and recurring collaborations.
- Monthly content addition trends reveal periods of increased content releases and platform activity.

Business Implications:

- Netflix can invest more in high-performing genres to maximize audience engagement and retention.
- Regional content strategies can be improved by focusing on countries with high content contribution and audience demand.
- Understanding yearly and monthly content trends can help optimize release schedules and promotional campaigns.
- Insights regarding dominant ratings and genres can support targeted marketing for different audience groups.
- Director productivity analysis can help identify valuable creative contributors for future collaborations and exclusive content production.
- Data-driven analytics using Big Data technologies enables streaming platforms to improve content planning, recommendation strategies, and business decision making.

CONCLUSION:

This project provided practical experience in implementing Big Data technologies using the Hadoop ecosystem on a real-world Netflix dataset. During the midterm phase, the focus was on setting up HDFS, uploading and organizing data, performing data cleaning and preprocessing using Apache Spark, and executing analytical queries through Apache Hive. These tasks helped in understanding distributed storage, parallel data processing, and SQL-based analysis on large datasets.

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In the final phase, the project was extended toward advanced analytics, including trend analysis, content segmentation, time-based aggregations, director productivity analysis, and country-wise genre analysis. HBase concepts were also explored by designing and storing summarized analytical metrics for fast retrieval. Various visualizations were created to better understand patterns and generate meaningful business insights. Overall, the project demonstrated how Hadoop, Spark, Hive, and HBase can work together to efficiently manage, process, and analyze large-scale data while supporting data-driven decision making.